MIL-A-85097/5A(AS)
17 October 1985
SUPERSEDING
MIL-A-85097/5(AS)
12 May 1978

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS, MARK 82 MOD 0 (MAN/SEAT SEPARATOR, LEFT) AND MARK 90 MOD 0 (MAN/SEAT SEPARATOR, RIGHT) (LOADED ASSEMBLIES)

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the motors described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to the Rocket Motors, Mark 82 Mod 0 and Mark 90 Mod 0, hereinafter referred to as the "rocket motors" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

STANDARDS

Military

MIL-STD-810 Environmental Test Methods

SPECIFICATIONS

Naval Air Systems Command (Code Ident 30003)

AS 3550 Propellant, N-43 (U)

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

944AS100 Rocket Motor, Mark 82 Mod 0 (Man/Seat Separator, Left)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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944AS117 Shipping Containers for Rocket Motors Mk 82 and Mk 90 Mods

970AS100 Rocket Motor, Mark 90 Mod 0 (Man/Seat Separator, Right)

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet, unless in direct conflict with the basic specification, shall be considered additional, not substitute. The Mark 82 Mod 0 Rocket Motor shall conform to Drawing 944AS100 and the Mark 90 Mod 0 Rocket Motor shall conform to Drawing 970AS100.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the requirements of table I.

TABLE I. Propellant N-43 (AS 3550).

Property	Temp (°F)	Zero time		7-Year Equivalent			
				0 % Rel. Humidity		22 % Rel. Humidity	
		Min	Max	Min	Max	Min	Max
S_m (maximum stress) psi	77	157	256	157	269	157	276
E_m (elongation at S_m) %	77	18	28	9	28	16	28
Y_m (Young's modulus) psi	77	1014	1946	1014	3094	1014	3114
B_r strand burn rate at 1000 psi) in/sec	77	0.60	0.67	0.60	0.71	0.60	0.71

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the requirements of AS 3550 when tested in accordance with MIL-A-85097.

3.3.2 Ballistic performance. The ballistic performance shall conform to table II.

TABLE II. Ballistic performance requirements.

Parameter	−40° ± 5°F		70° ± 5°F		160° ± 5°F	
	Min	Max	Min	Max	Min	Max
Ignition delay (ms)	-	10	-	10	-	10
Action time (ms)	-	263[1]	-	235	-	212
Rocket motor resultant thrust (max)	439[1]	725	510	840	570	950[1]
Rocket motor impulse (lb-s)	97[1]	-	97[1]	-	97[1]	-
Time to 375-pound thrust (ms)	-	26[1]	-	26[1]	-	26[1]
Burn time (ms)	163	-	140	-	120[1]	-

[1] System requirements.

3.3.3 Finish. Color shall be dark gull gray, No. 36231 in accordance with FED-STD-595.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Examinations and tests.

4.1.1 Preproduction.

4.1.1.1 Environmental.

4.1.1.1.1 Vibration. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification. Two rocket motors shall be subjected to vibration testing. One rocket motor shall be temperature conditioned for not less than 4 hours at $-40° \pm 5°F$ and the other at $160° \pm 5°F$. The rocket motors shall be maintained at specified conditioning temperature during these tests. If the rocket motor must be removed from the conditioning chamber or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes for every minute the rocket motor spent outside the specified temperature envelope shall be required, up to a maximum of 4 hours. A continuous temperature history of the rocket motor shall be kept for all conditioning and test time. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise specified in the contract.

4.1.1.1.1.1 Resonance survey. Resonance surveys shall be conducted on all loaded rocket motors in three axes at the maximum and minimum conditioning temperatures. If two or more rocket motors are to be vibrated simultaneously, the fixture configuration, accelerometer location, and individual rocket motor input acceleration tolerance shall be approved by the Government procuring activity. Input vibratory acceleration shall be in accordance with curve J, figure 514.2-2 of MIL-STD-810. All resonances shall be as noted. During the resonance survey a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on an instrument acceptable to the Government and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

a. A resonance is considered as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) exceeds a value of 1.5 to 1.

b. If the resonance survey indicates a frequency band wherein the amplitude ratio exceeds 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered an additional resonant point.

4.1.1.1.1.2 Endurance vibration. Rocket motors specified shall be vibrated for 24 hours, 8 hours in each of the three axes. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant survey. Input vibratory acceleration for each resonant frequency shall be in accordance with the acceleration and displacement of curve J, figure 514.2-2 of MIL-STD-810. If no resonant frequency is detected in an axis, the rocket motor shall be vibrated in accordance with curve J, figure 514.2-2 of MIL-STD-810. Total vibration time for the rocket motors shall be 24 hours which shall include resonant and nonresonant frequencies, as applicable. Endurance vibration of frequency bands which are classified as resonances in 4.1.1.1.1.1 shall be conducted as continuous repetitive sweeps across the entire band up in frequency and return at a rate not greater than 0.25 octave per minute.

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4.2 **Preproduction/production.**

4.2.1 **Conditioning temperatures.** The conditioning temperatures and tolerance shall be as follows:

a. Cold: −40° ± 5°F
b. Hot: 160° ± 5°F

4.2.2 **Conditioning period.** Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: not less than 5 hours
b. Temperature cycling: not less than 5 hours

5. PACKAGING

5.1 **Packing.** Rocket motors shall be packed as specified on Drawing 944AS117.

6. NOTES

6.1 **Intended use.** The Rocket Motor Mk 82 Mod 0 is intended for use as a means of insuring positive separation of an aircrewman from his ejection seat for the S-3, A-4, and A-7 aircraft escape systems. The Rocket Motor Mk 90 Mod 0 is used only in the S-3 aircraft.

6.2 **Changes from previous issue.** Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/5A(AS)	(AEPS); Rocket Motors, MARK 82 MOD 0

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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